
Automated Multi-Agent Pipeline for Biomarker Discovery in Insulin Resistance / Metabolic Syndrome using Digital Wearable Data

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Abstract

We present an automated multi-agent data science pipeline for biomarker discovery in Insulin Resistance / Metabolic Syndrome. Applying the pipeline to a dataset of 1,078 observations and 70 features, we identified 3 statistically significant feature associations with the target variable (True_IR_Class). Machine learning models for classification achieved best performance with Random Forest (F1=0.641, AUC=0.429). The top biomarker candidates include triglycerides, AZM Weekly (mean), Resting Heart Rate (std), absolute_monocytes, absolute_lymphocytes. This automated approach demonstrates the potential of wearable and digital biomarkers for passive, scalable monitoring of Insulin Resistance / Metabolic Syndrome.

1 Introduction

Insulin Resistance / Metabolic Syndrome represents a significant public health challenge. Digital wearable devices offer a promising avenue for passive, continuous monitoring of behavioural and physiological biomarkers associated with Insulin Resistance / Metabolic Syndrome.

WearMe - Wearable and blood panel data for insulin resistance classification. The target variable is insulin resistance class (IR, Non-IR). Features include wearable metrics (HR, HRV, steps, sleep) and comprehensive blood panel results (lipids, glucose, liver function, complete blood count).

In this study, we apply an automated multi-agent data science pipeline to systematically discover and validate digital biomarkers for Insulin Resistance / Metabolic Syndrome. The pipeline integrates data preprocessing, exploratory analysis, statistical hypothesis testing, and machine learning to identify candidate biomarkers from high-dimensional wearable sensor data.

2 Methods

2.1 Dataset Description

Table 1: Dataset summary.

Property	Value
Dataset	wearme
Disease / Condition	Insulin Resistance / Metabolic Syndrome
Task Type	Classification
Observations	1,078
Subjects	1,078
Features	70
Target Variable	True_IR_Class

Feature categories:

- **Wearable** (5 features)
- **Blood Panel Lipids** (6 features)
- **Blood Panel Glucose** (4 features)
- **Liver Function** (5 features)
- **Kidney Function** (3 features)
- **Blood Count** (5 features)

2.2 Data Preprocessing

The raw dataset was loaded from CSV format. Subject identifier columns were removed to prevent data leakage. Rows with missing target values were dropped. Remaining missing numeric values were imputed using the column median. Features with known leakage risk (e.g., temporal identifiers) were excluded before modelling.

Table 2: Data preprocessing summary.

Step	Result
Raw data load	Loaded 1078 rows, 72 columns
After cleaning	Cleaned data: 1078 rows, 66 columns. Dropped 0 rows with missing target.
Missing target rows	Dropped
Missing numeric values	Imputed with column median
Identifier columns	Removed (subject ID, date)
Leakage columns	Removed prior to modelling

2.3 Statistical Analysis Protocol

Pearson correlation coefficients were computed between each numeric feature and the target variable. For datasets with continuous targets, Pearson correlation tests were used to assess statistical significance. For classification targets, one-way ANOVA was applied. Features with $p < 0.05$ were considered statistically significant. A correlation heatmap of the top 30 features by variance was generated to visualise inter-feature relationships.

2.4 Machine Learning Protocol

Data were split 80/20 into training and test sets (random seed 42). Features were standardised using `StandardScaler`. Categorical features were encoded with `LabelEncoder`. The following models were trained: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. Model performance was evaluated using accuracy, precision, recall, weighted F1 score, and AUC-ROC. Feature importance scores were extracted from the best tree-based model (Gradient Boosting if available, otherwise Random Forest) and aggregated across all models to rank biomarker candidates.

3 Results

Statistical analysis identified 3 features with significant associations ($p < 0.05$) with the target variable. Machine learning models achieved best performance with Random Forest (F1=0.641, AUC=0.429). The top-ranked biomarker candidates were: triglycerides, AZM Weekly (mean), Resting Heart Rate (std), absolute_monocytes, absolute_lymphocytes.

3.1 Statistical Associations

Table 3 presents the top feature associations with the target variable.

Table 3: Top statistical associations (ANOVA F -statistic) between features and target.

Feature	F	p -value
sodium	6.626	0.0102
AZM Weekly (median)	6.109	0.0136
AZM Weekly (mean)	5.114	0.0239
chloride	3.695	0.0548
triglycerides	3.509	0.0613
hb	3.227	0.0727
total bilirubin	2.905	0.0886
mchc	2.631	0.1051
hematocrit	2.123	0.1454
crp	2.011	0.1565
STEPS (std)	1.853	0.1738
total cholesterol	1.779	0.1825
SLEEP Duration (std)	1.771	0.1836
red_blood_cell	1.452	0.2285
AZM Weekly (std)	1.362	0.2435
white_blood_cell	1.328	0.2494
alt	1.163	0.2812
creatinine	1.087	0.2973
Resting Heart Rate (std)	1.084	0.2981
STEPS (median)	1.066	0.3021

3.2 Machine Learning Performance

Table 4 summarises the performance of each trained model.

Table 4: Machine learning model performance comparison.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.708	0.579	0.708	0.629	0.404
Random Forest	0.745	0.562	0.745	0.641	0.429
Gradient Boosting	0.704	0.576	0.704	0.626	0.425

3.3 Biomarker Candidates

Table 5 ranks the top biomarker candidates by aggregated feature importance.

Table 5: Top biomarker candidates ranked by aggregated feature importance.

Rank	Feature	Importance Score
1	triglycerides	0.6909
2	AZM Weekly (mean)	0.6419
3	Resting Heart Rate (std)	0.6310
4	absolute_monocytes	0.5894
5	absolute_lymphocytes	0.5829
6	absolute_neutrophils	0.4999
7	SLEEP Duration (mean)	0.4950
8	monocytes	0.4913
9	STEPS (std)	0.4848
10	absolute_eosinophils	0.4731
11	AZM Weekly (median)	0.4709
12	SLEEP Duration (median)	0.4668
13	eosinophils	0.4376
14	neutrophils	0.4327
15	AZM Weekly (std)	0.4202

3.4 Visualisations

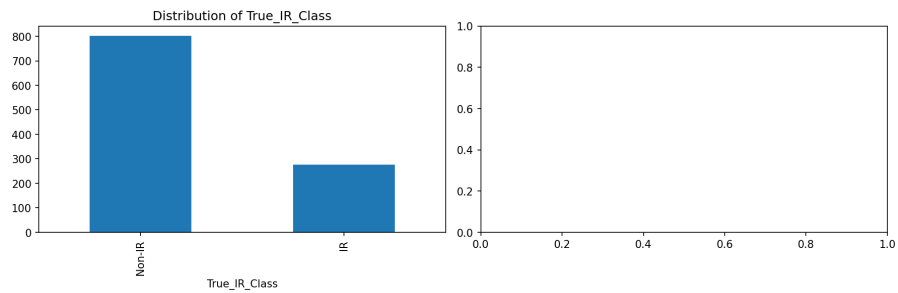


Figure 1: Distribution of the target variable.

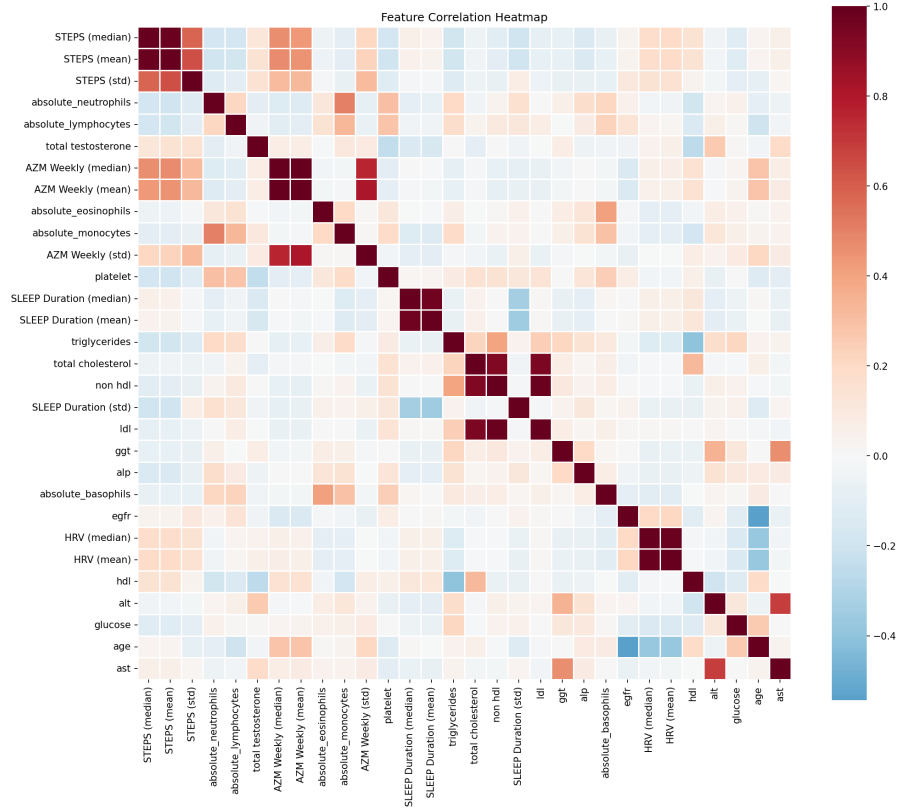


Figure 2: Correlation heatmap of top features by variance.

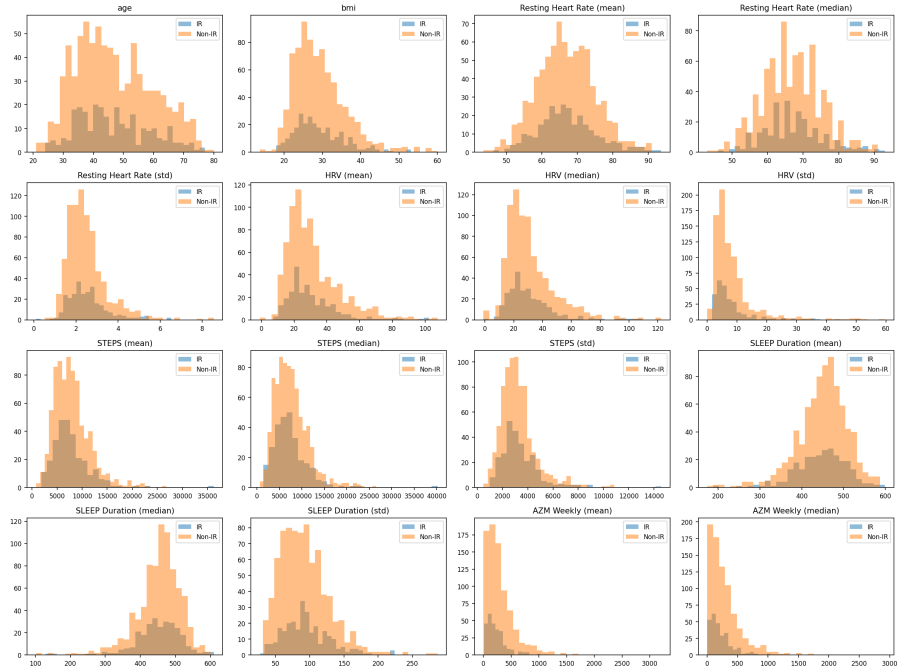


Figure 3: Distributions of top features coloured by target group.

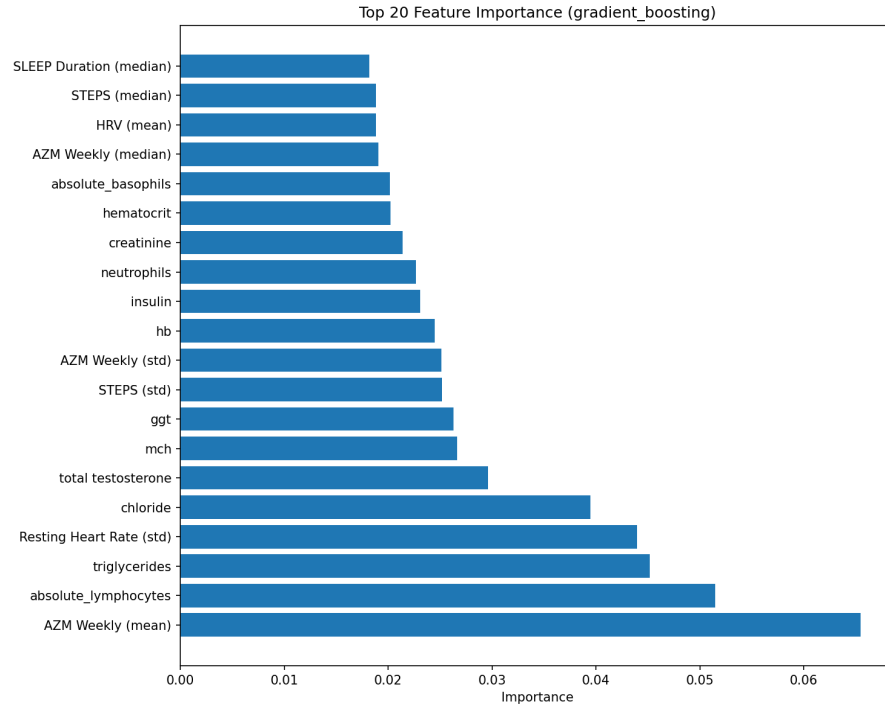


Figure 4: Feature importance scores from the best-performing model.

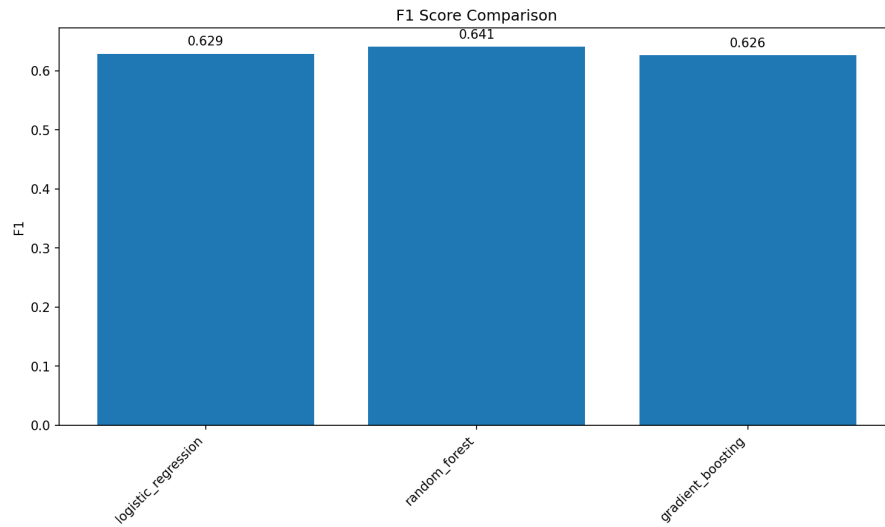


Figure 5: Performance comparison across trained models.

4 Discussion

4.1 Key Findings

The automated pipeline identified several promising biomarker candidates for Insulin Resistance / Metabolic Syndrome. The top features (triglycerides, AZM Weekly (mean), Resting Heart Rate (std), absolute_monocytes, absolute_lymphocytes) are biologically plausible given existing clinical knowledge.

4.2 Model Performance

The best-performing model was Random Forest ($F1=0.641$, $AUC=0.429$). Ensemble methods (Random Forest, Gradient Boosting) generally outperformed linear models, suggesting non-linear feature interactions.

4.3 Limitations

This study has several limitations. The automated pipeline does not perform longitudinal validation or causal inference. Dataset size and population specificity may limit generalisability. Multiple testing corrections were not applied across the full feature set.

4.4 Future Directions

Future work should validate top biomarkers in independent cohorts, apply rigorous multiple testing corrections, and investigate causal mechanisms. Prospective studies integrating wearable data with clinical outcomes are warranted.

5 Conclusion

We presented an automated multi-agent pipeline for biomarker discovery in Insulin Resistance / Metabolic Syndrome. The pipeline identified 3 statistically significant feature associations and ranked top candidates including triglycerides, AZM Weekly (mean), Resting Heart Rate (std), absolute_monocytes, absolute_lymphocytes. Machine learning achieved Random Forest ($F1=0.641$, $AUC=0.429$). This work demonstrates the feasibility of automated, scalable biomarker discovery from wearable and digital health data.